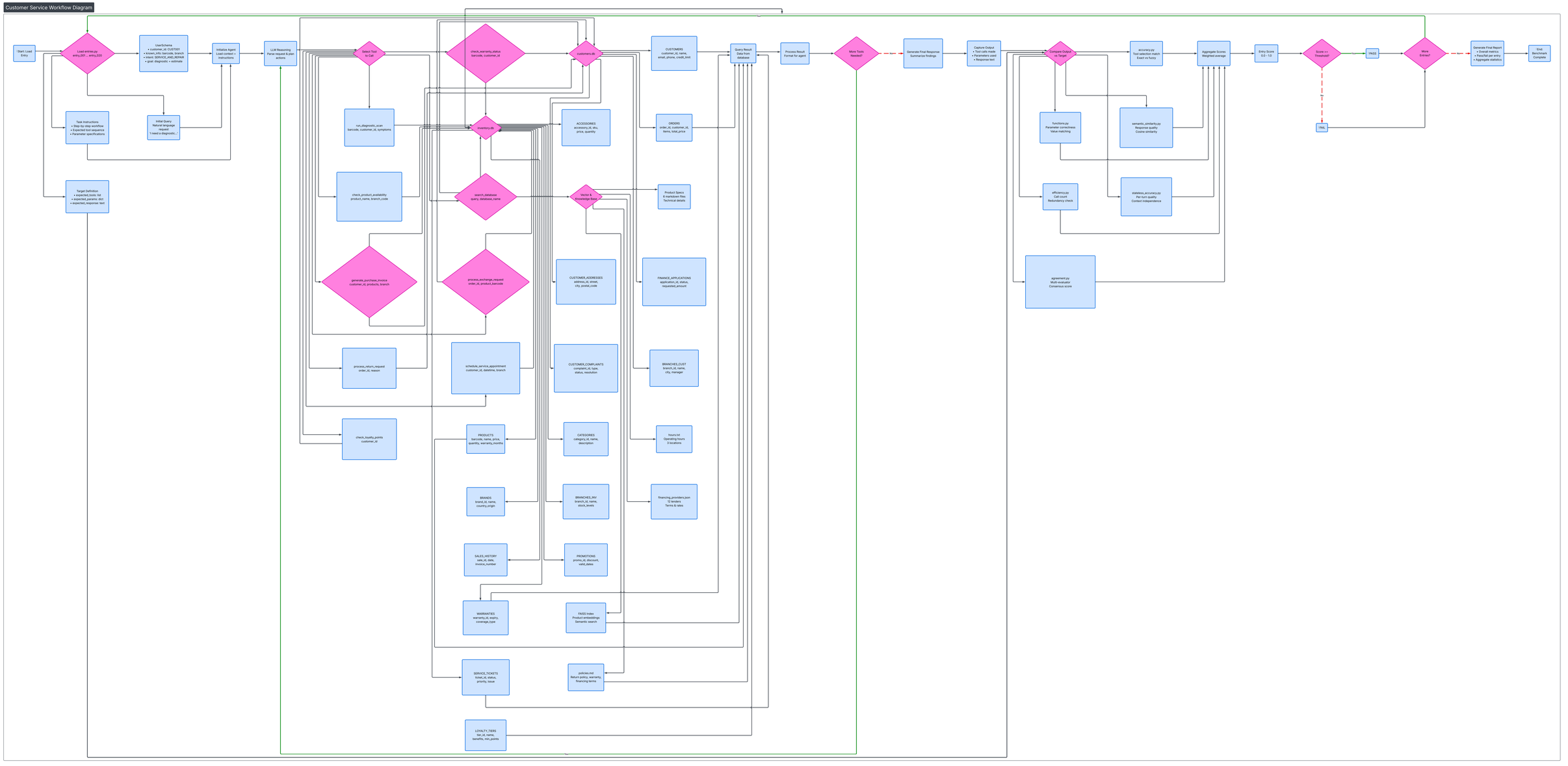
Customer Service Workflow Diagram
Start: Load Entry
Load entries.py
entry_001 ... entry_020
UserSchema
• customer_id: CUST001
• known_info: barcode, branch
• intent: SERVICE_AND_REPAIR
• goal: diagnostic + estimate
Initialize Agent
Load context + instructions
LLM Reasoning
Parse request & plan actions
Task Instructions
• Step-by-step workflow
• Expected tool sequence
• Parameter specifications
Initial Query
Natural language request
"I need a diagnostic..."
Target Definition
• expected_tools: list
• expected_params: dict
• expected_response: text
Select Tool to Call
run_diagnostic_scan
barcode, customer_id, symptoms
check_product_availability
product_name, branch_code
generate_purchase_invoice
customer_id, products, branch
process_return_request
order_id, reason
check_loyalty_points
customer_id
check_warranty_status
barcode, customer_id
inventory.db
search_database
query, database_name
process_exchange_request
order_id, product_barcode
schedule_service_appointment
customer_id, datetime, branch
customers.db
ACCESSORIES
accessory_id, sku,
price, quantity
Vector & Knowledge Base
CUSTOMER_ADDRESSES
address_id, street,
city, postal_code
CUSTOMER_COMPLAINTS
complaint_id, type,
status, resolution
PRODUCTS
barcode, name, price,
quantity, warranty_months
CATEGORIES
category_id, name,
description
BRANDS
brand_id, name,
country_origin
BRANCHES_INV
branch_id, name,
stock_levels
SALES_HISTORY
sale_id, date,
invoice_number
PROMOTIONS
promo_id, discount,
valid_dates
WARRANTIES
warranty_id, expiry,
coverage_type
FAISS Index
Product embeddings
Semantic search
SERVICE_TICKETS
ticket_id, status,
priority, issue
policies.md
Return policy, warranty,
financing terms
LOYALTY_TIERS
tier_id, name,
benefits, min_points
CUSTOMERS
customer_id, name,
email, phone, credit_limit
ORDERS
order_id, customer_id,
items, total_price
Product Specs
6 markdown files
Technical details
FINANCE_APPLICATIONS
application_id, status,
requested_amount
BRANCHES_CUST
branch_id, name,
city, manager
hours.txt
Operating hours
3 locations
financing_providers.json
12 lenders
Terms & rates
Query Result
Data from database
Process Result
Format for agent
More Tools Needed?
Generate Final Response
Summarize findings
Capture Output
• Tool calls made
• Parameters used
• Response text
Compare Output vs Target
functions.py
Parameter correctness
Value matching
efficiency.py
Call count
Redundancy check
agreement.py
Multi-evaluator
Consensus score
semantic_similarity.py
Response quality
Cosine similarity
stateless_accuracy.py
Per-turn quality
Context independence
accuracy.py
Tool selection match
Exact vs fuzzy
Aggregate Scores
Weighted average
Entry Score
0.0 - 1.0
Score >= Threshold?
PASS
FAIL
More Entries?
Generate Final Report
• Overall metrics
• Pass/fail per entry
• Aggregate statistics
End: Benchmark Complete